CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

# 1.2 SI Units

**Syllabus Section:** 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

## Question 1 Easy [1 mark]

Which of the following is an SI base unit?

**A** Current

**B** Gram

**C** Kelvin

**D** Volt

### Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The main SI units you need to be aware of are:
  - Metre (m)
  - Kilogram (kg)
  - Second (s)
  - Ampere (A)
  - Kelvin (K)
  - Mole (mol)
- Kelvin is the only SI unit listed

**A** is incorrect as current is a quantity, not a unit

**B** is incorrect as the correct SI unit for mass is the kilogram

**D** is incorrect as the volt is a derived unit

**Question 2** Easy [1 mark]

Five energies are listed below.

5kJ

5mJ

5MJ

5nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

**A** 5kJ → 5mJ → 5MJ → 5nJ

**B** 5nJ → 5kJ → 5MJ → 5mJ

**C** 5nJ → 5mJ → 5kJ → 5MJ

**D** 5mJ → 5nJ → 5kJ → 5MJ

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- If we convert each power of ten we get:

| | | |
|---|---|---|
| • 5 kJ | = | $5 \times 10^{3}$ J |
| • 5 mJ | = | $5 \times 10^{-3}$ J |
| • 5 MJ | = | $5 \times 10^{6}$ J |
| • 5 nJ | = | $5 \times 10^{-9}$ J |

- Therefore, from smallest to largest: 5nJ → 5mJ → 5kJ → 5MJ

**Question 3** Easy [1 mark]

Which product-pair of metric prefixes has the greatest magnitude?

A pico × mega

B nano × kilo

C micro × giga

D milli × tera

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- milli × tera is equal to $10^{-3} \times 10^{12} = 10^{9}$

**A** is incorrect as this is equal to $10^{-12} \times 10^{6} = 10^{-6}$

**B** is incorrect as this is equal to $10^{-9} \times 10^{3} = 10^{-6}$

**C** is incorrect as this is equal to $10^{-6} \times 10^{9} = 10^{3}$

**Question 4** Easy [1 mark]

Which statement includes a correct unit?

**A** Energy = 7.8 N s

**B** Force = 3.8 N s

**C** Momentum = 6.2 N s

**D** Torque = 4.7 N s

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Change in momentum (or impulse) is equal to force × time, giving a unit of N s

**A** is incorrect as the unit of energy is J or kg $m^2$ $s^{-2}$ or N m

**B** is incorrect as the unit of force is N or kg m $s^{-2}$

**D** is incorrect as the unit of torque is N m or kg $m^2$ $s^{-2}$

**Question 5** Easy [1 mark]

Which quantity can be measured in electronvolts (eV)?

**A** Electric charge

**B** Electric potential

**C** Energy

**D** Power

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The definition of an electron volt is the energy that an electron gains when it travels through a potential of one volt
- 1 eV is equal to $1.6 \times 10^{-19}$ J

**Question 6** Easy [1 mark]

Which of the following is not a SI base unit?

**A** kilogram

**B** metre

**C** second

**D** volt

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The volt is a derived unit (1 V = 1 kg $m^2$ $s^{-3}$ $A^{-1}$ )

**A** is incorrect because kilogram is a base unit

**B** is incorrect because metre is a base unit

**C** is incorrect because second is a base unit

The base units you need to be familiar with are:

- Mass - kilogram (kg)
- Length - metre (m)
- Time - second (s)
- Temperature - Kelvin (K)
- Electric current - ampere (A)

**Question 7** Easy [1 mark]

Which product pair of metric prefixes has the smallest magnitude?

**A** nano × mega

**B** tera × pico

**C** centi × kilo

**D** micro × giga

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- nano = $1 \times 10^{-9}$
- mega = $1 \times 10^{6}$
- nano × mega = $(1 \times 10^{-9}) \times (1 \times 10^{6}) = (1 \times 10^{-3}) = 0.001$
- This is the smallest answer

**B** is incorrect because:

- tera = $1 \times 10^{12}$
- pico = $1 \times 10^{-12}$
- tera × pico = $(1 \times 10^{12}) \times (1 \times 10^{-12}) = (1 \times 10^{0}) = 1$
- This is greater than **A**

**C** is incorrect because:

- centi = $1 \times 10^{-2}$
- kilo = $1 \times 10^{3}$
- centi × kilo = $(1 \times 10^{-2}) \times (1 \times 10^{3}) = (1 \times 10^{1}) = 10$
- This is greater than **A**

**D** is incorrect because:

- micro = $1 \times 10^{-6}$
- giga = $1 \times 10^{9}$
- micro × giga = $(1 \times 10^{-6}) \times (1 \times 10^{9}) = (1 \times 10^{3}) = 1000$

- This is greater than **A**

You are expected to know the following prefixes:

- pico (p)
- nano (n)
- micro (μ)
- milli (m)
- centi (c)
- deci (d)
- kilo (k)
- mega (M)
- giga (G)
- tera (T)

**Question 8** Easy [1 mark]

Which unit is **not** an SI base unit?

**A** A

**B** K

**C** mol

**D** N

**Mark Scheme & Explanation:**

*The correct answer is D*

- The newton (N) is a derived unit (1 N = 1 kg m $s^{-2}$)

**A** is incorrect because ampere (A) is an SI base unit

**B** is incorrect because kelvin (K) is an SI base unit

**C** is incorrect because mole (mol) is an SI base unit

The derived units you need to be familiar with are:

- force - newton (N)
- energy - joule (J)
- power - watt (W)
- pressure - pascal (Pa)
- potential difference - volt (V)

**Question 9** Medium [1 mark]

The average kinetic energy *E* of a gas molecule is given by the equation

$$E = \frac{3}{2}kT$$

where *T* is the thermodynamic (kelvin) temperature.

What are the SI base units of *k*?

**A** $\mathrm{kg^{-1}\ m^{-1}\ s^{2}\ K}$

**B** $\mathrm{kg^{-1}\ m^{-2}\ s^{2}\ K}$

**C** $\mathrm{kg\ m\ s^{-2}\ K^{-1}}$

**D** $\mathrm{kg\ m^{2}\ s^{-2}\ K^{-1}}$

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- We can ignore any numbers in the formula since they do not have any units.
- $k \propto \frac{E}{T}$ so, the unit will be along the lines of $\mathrm{J\ K^{-1}}$
- We can reduce J to its SI unit by considering the equation for kinetic energy $E_k \propto mv^2$
  - $\mathrm{J = kg\ m\ s^{-1\,2} = kg\ m^{2}\ s^{-2}}$
- The units of *k* are
  - $\mathrm{J\ K^{-1} = kg\ m^{2}\ s^{-2}\ K^{-1}}$

**Question 10** Medium [1 mark]

The units of all physical quantities can be expressed in terms of SI base units.

Which pair contains quantities with the same base units?

**A** Force and momentum

**B** Pressure and Young modulus

**C** Power and kinetic energy

**D** Mass and weight

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Pressure is equal to
  - $P = \frac{F}{A} = \mathrm{Nm^{-2}}$
- Young Modulus is equal to
  - $Y = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress: $\sigma = \frac{F}{A}$
  - Strain: $\varepsilon = \frac{x}{l}$
  - $Y = \frac{\sigma}{\varepsilon} = \frac{F}{A} \times \frac{l}{x} = \frac{\mathrm{N}}{\mathrm{m^2}} \times \frac{\mathrm{m}}{\mathrm{m}} = \mathrm{Nm^{-2}}$

Therefore, pressure and Young Modulus have the same base units.

**A** is incorrect as the base unit for force is kg m $s^{-2}$ and the base unit for momentum is kg m $s^{-1}$

**C** is incorrect as the unit for power is J $s^{-1}$ and the unit for energy is J

**D** is incorrect as The base unit for mass is kg and the unit for weight is N or kg m $s^{-2}$

## Question 11 Medium [1 mark]

The table contains some quantities, together with their symbols and units.

| Quantity | Symbol | Unit |
|---|---|---|
| Gravitational field strength | $g$ | $\text{N kg}^{-1}$ |
| Density of liquid | $\rho$ | $\text{kg m}^{-3}$ |
| Vertical height | $h$ | m |
| Volume of part of liquid | $V$ | $\text{m}^3$ |

Which expression has the units of energy?

**A** $\rho g^2 h$

**B** $\frac{\rho h V}{g}$

**C** $\frac{\rho g}{hV}$

**D** $g \rho h V$

### Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The base unit of energy is $\text{J} = \text{kgms}^{-1\,2} = \text{kg m}^2 \text{ s}^{-2}$
  - $g\rho hV = \text{ms}^{-2}\,\text{kg m}^{-3}\,\text{mm}^3$
    $= \text{N m}$
    $= \text{kg m s}^{-2}\,\text{m}$
    $= \text{kg m}^2 \text{ s}^2$
- Therefore, $g \rho h V$ has the same units as energy

**A** is incorrect as the units of $\rho g^2 h = \text{kg m}^{-3}\,\text{m s}^{-2\,2}\,\text{m} = \text{kg s}^{-4}$

**B** is incorrect as the units of $\frac{\rho h V}{g} = \text{kg m}^{-3}\,\text{mm}^3\,\text{m s}^{-2\,-1} = \text{kg s}^2$

**C** is incorrect as the units of $\frac{\rho g}{\hbar V} = \text{kg m}^{-3}\,\text{m s}^{-2}\,\text{m}^{-1}\,\text{m}^{3\,-1} = \text{kg m}^{-6}\,\text{s}^{-2}$

## Question 12 Medium [1 mark]

What is the ratio $\frac{10^{-3}\ \text{THz}}{10^{3}\ \text{kHz}}$ ?

**A** $10^{-9}$

**B** $10^{-6}$

**C** $10^{0}$

**D** $10^{3}$

### Mark Scheme & Explanation:

*The correct answer is **D** because:*

- T = tera = $10^{12}$
  - $10^{-3} \times 10^{12} = 10^{12\,-\,3} = 10^{9}$
- k = kilo = $10^{3}$
  - $10^{3} \times 10^{3} = 10^{3\,+\,3} = 10^{6}$
- $\frac{10^{9}}{10^{6}} = 10^{9\,-\,6} = 10^{3}$

**Question 13** Medium [1 mark]

Which of the following correctly expresses the volt in terms of SI base units?

**A** $A\Omega$

**B** $WA^{-1}$

**C** $kg\ m^2\ s^{-1}\ A^{-1}$

**D** $kg\ m^2\ s^{-3}\ A^{-1}$

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- Potential difference is equal to $\frac{work\ done}{charge}$
- So, the unit of the volt is J $C^{-1}$
- The base unit of energy $E = \frac{1}{2}mv^2$ is
  - $J = kgms^{-1}{}^2 = kgm^2s^{-2}$
- The base unit of charge $Q = It$ is
  - $C = As$
- So, the base unit of the volt is
  - $J\ C^{-1} = kg\ m^2\ s^{-2}\ A^{-1}\ s^{-1}$
- $V = kg\ m^2\ s^{-3}\ A^{-1}$

**Question 14** Medium [1 mark]

The equation relating pressure and density is $P = \rho g h$

How can both sides of this equation be written in terms of base units?

**A** $\mathrm{N\ m^{-1} = kg\ m^{-3}\,m\ s^{-1}\,m}$

**B** $\mathrm{N\ m^{-2} = kg\ m^{-2}\,m\ s^{-2}\,m}$

**C** $\mathrm{kg\ m^{-1}\ s^{-2} = kg\ m^{-3}\,m\ s^{-2}\,m}$

**D** $\mathrm{kg\ m^{-1}\ s^{-1} = kg\ m^{-1}\,m\ s^{-2}\,m}$

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Pressure is equal to
  - $P = \frac{F}{A} = \mathrm{N\ m^{-2} = kg\ ms^{-2}m^{-2} = kg\ m^{-1}\ s^{-2}}$
- On the RHS we can combine units to give
  - $\mathrm{kg\ m^{-3}mm\ s^{-2} = kg\ m^{-1}\ s^{-2}}$
- The LHS matches the RHS so the correct answer is C

**A** is incorrect as the $\mathrm{N\ m^{-1}}$ on the LHS is the incorrect unit for pressure

**B** is incorrect as the units are correct on the LHS, but on the RHS $\mathrm{kg\ m^{-2}\,m\ s^{-2}\,m}$ simplifies to $\mathrm{kg\ s^{-1}}$ which is incorrect

**D** is incorrect as the $\mathrm{kg\ m^{-1}\ s^{-1}}$ on the LHS is the incorrect base unit for pressure

**Question 15** Medium [1 mark]

What is the ratio $\frac{10^2\ \text{THz}}{10^{-4}\ \text{pHz}}$ ?

**A** $10^{-30}$

**B** $10^{-2}$

**C** $10^{2}$

**D** $10^{30}$

**Mark Scheme & Explanation:**

*The correct answer is D*

- $T = \text{tera} = 10^{12}$

$$10^{2} \times 10^{12} = 10^{(12+2)} = 10^{14}$$

- $p = \text{pico} = 10^{-12}$

$$10^{-4} \times 10^{-12} = 10^{(-12-4)} = 10^{-16}$$

- Determine the ratio

$$\frac{10^{14}}{10^{-16}} = 10^{14--16} = 10^{30}$$

- This is answer **D**

**Question 16** Hard [1 mark]

The speed $v$ of a liquid leaving a tube depends on the change in pressure $\Delta P$ and the density $\rho$ of the liquid. The speed is given by the equation

$$v = k\frac{\Delta P}{\rho}^{n}$$

where $k$ is a constant that has no units.

What is the value of $n$?

**A** $\frac{1}{2}$

**B** 1

**C** $\frac{3}{2}$

**D** 2

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- Combining $\Delta P = \frac{F}{A}$ and $F = ma$
  - $\Delta P = \frac{ma}{A} = \frac{\text{kg ms}^{-2}}{\text{m}^2} = \text{kg m}^{-1}\text{ s}^{-2}$
- The units of density are
  - $\rho = \frac{m}{v} = \text{kg m}^{-3}$
  - $\frac{\Delta P}{\rho} = \frac{\text{kg m}^{-1}\text{ s}^{-2}}{\text{kg m}^{-3}} = \text{m}^3\text{ m}^{-1}\text{ s}^{-2} = \text{m}^2\text{ s}^{-2}$
  - $\text{m s}^{-1} = \text{m}^2\text{ s}^{-2}{}^{n}$
- Therefore $n = \frac{1}{2}$

**Question 17** Hard [1 mark]

The frictional force *F* on a sphere falling through a fluid is given by the formula

$$F = 6\pi r\eta v$$

where *r* is the radius of the sphere, $\eta$ is a constant relating to the fluid and *v* is the velocity of the sphere.

What are the units of $\eta$ ?

**A** $\text{kg m s}^{-1}$

**B** $\text{kg m}^{-1} \text{ s}^{-1}$

**C** $\text{kg m s}^{-3}$

**D** $\text{kg m}^{3} \text{ s}^{-3}$

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Once any constants are removed, $F \propto r\eta v$
  - $\eta \propto \frac{F}{rv}$
- $\eta = \text{N m}^{-1}\text{m s}^{-1\,-1}$
- $\eta = \text{kg m s}^{-2} \text{ m}^{-2} \text{ s}$
- $\eta = \text{kg m}^{-1} \text{ s}^{-1}$
- Therefore, the units for $\eta$ are $\text{kg m}^{-1} \text{ s}^{-1}$

**Question 18** Hard [1 mark]

When a constant braking force is applied to a vehicle moving at speed *v*, the distance *d* moved by the vehicle as it comes to rest is given by the expression

$$d = kv^2$$

where *k* is a constant.

When *d* is measured in metres and *v* is measured in metres per second, the constant has a value of $k_1$.

What is the value of the constant *k* when the distance is measured in metres, and the speed is measured in kilometres per hour?

**A** $0.077k_1$

**B** $0.278k_1$

**C** $3.60k_1$

**D** $13.0k_1$

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- When $d = k_1 v^2$ the unit of $k_1$ is
  - $\frac{d}{v^2} = \text{mm s}^{-1} = \text{m}^{-1}\ \text{s}^2$
- When $d = kv^2$ the unit of *k* is
  - $\frac{d}{v^2}\text{mkm h}^{-1} = \text{mm s}^{-1\ -2}\ \frac{1000}{60\times 60}^{-2} = 13.0\ \text{m}^{-1}\ \text{s}^2$
- Comparing the two units we can see $k_1$ = 13.0*k*, so *k* = 0.077$k_1$

**B** and **C** are incorrect because these are the answers you get if you don't square the conversion i.e. $\frac{3000}{1000} = 3.6$

**D** is incorrect because $k_1$ is equal to 13.0*k*, not the other way around

**Question 19** Hard [1 mark]

The drag coefficient $C_d$ is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where $F$ is the drag force on the car, $\rho$ is the density of the air, $A$ is the cross-sectional area of the car and $v$ is the speed of the car.

What is the value of $n$?

A 1

B 2

C 3

D 4

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Since $C_d$ is a quantity with no units we can disregard it for this question
- That leaves us with F $\alpha$ $\rho v^n A$
- So, $v^n$ $\alpha$ $\frac{F}{\rho A}$
- $\mathrm{ms^{-1}}^n = \mathrm{Nkg\ m^{-3}}^{-1}\mathrm{m^2}^{-1}$

  $= \mathrm{kg\ m\ s^{-2}\,kg^{-1}\,m^3\,m^{-2}}$

  $= \mathrm{m^2\,s^{-2}}$
- Comparing $\mathrm{m\ s^{-1}}^n = \mathrm{m^2\,s^{-2}}$ we can see that $n = 2$

**Question 20** Hard [1 mark]

Which formula could be correct for the speed $v$ of ocean waves in terms of the density $\rho$ of sea-water, the acceleration of free fall g, the height $h$ of the wave and the wavelength $\lambda$?

A $v = \sqrt{\rho g h}$

B $v = \sqrt{\frac{g}{h}}$

C $v = \sqrt{g\lambda}$

D $v = \sqrt{\frac{g}{\rho}}$

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The unit of
  - $v$ is $\mathrm{m\ s^{-1}}$
  - $g$ is $\mathrm{m\ s^{-2}}$
  - $\lambda$ is $\mathrm{m}$
- The unit of $g\lambda$ is
  - $\mathrm{m\ s^{-2}\,m^{\frac{1}{2}} = m^2 s^{-2\,\frac{1}{2}} = m\ s^{-1}}$
- The units of the LHS and the RHS match so the correct answer is C

**A** is incorrect because the unit of $\sqrt{\rho g h}$ is $\mathrm{kg\ m^{-3}\,m\ s^{-2}\,m^{\frac{1}{2}} = kg\ m^{-1}\,s^{-2\,\frac{1}{2}}}$

**B** is incorrect because the unit of $\sqrt{\frac{g}{h}}$ is $\mathrm{m\ s^{-2}\,m^{-1\,\frac{1}{2}} = s^{-1}}$

**D** is incorrect because the unit of $\sqrt{\frac{g}{\rho}}$ is $\mathrm{m\ s^{-2}\,kg^{-1}\,m^{3\,\frac{1}{2}} = kg^{-1}\,m^4\,s^{-2\,\frac{1}{2}}}$

**Question 21** Hard [1 mark]

The equation for the terminal velocity of a sphere of radius *r* falling under acceleration due to gravity, *g*, through a liquid with density $\rho$ and viscosity $\eta$, measured in kg $m^{-1}$ $s^{-1}$ is:

$$v_t = \frac{2}{9}\frac{gr^n\rho}{\eta}$$

What is the value of *n*?

**A** 0.5

**B** 1

**C** 2

**D** 3

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The units of the variables are:
  - $g$ : m $s^{-2}$
  - $r$ : m
  - $\rho$ : kg $m^{-3}$
  - $\eta$ : kg $m^{-1}$ $s^{-1}$
- Substituting these units into the equation and ignoring constants:
  - $\mathrm{m\ s^{-1}} = \frac{(\mathrm{m\ s^{-2}})(\mathrm{m^n})(\mathrm{kgm^{-3}})}{\mathrm{kg\ m^{-1}\ s^{-1}}}$
  - $\mathrm{m\ s^{-1}} = \frac{\mathrm{m^{-2}\ s^{-2}\ m^n\ kg}}{\mathrm{kg\ m^{-1}\ s^{-1}}}$
  - $\mathrm{m\ s^{-1}} = \mathrm{m^{-1}\ s^{-1}\ m^n}$
- For the sides to balance *n* must be 2 so that:
  - $\mathrm{m\ s^{-1}} = \mathrm{m^{-1}\ s^{-1}\ m^2}$
  - $\mathrm{m\ s^{-1}} = \mathrm{m\ s^{-1}}$

**A** is incorrect because:

- $\mathrm{m\ s^{-1}} = \frac{\mathrm{m\ s^{-2}\,(m^{\frac{1}{2}})(kg\ m^{-3})}}{\mathrm{kg\ m^{-1}\ s^{-1}}}$
- $\mathrm{m\ s^{-1}} = \mathrm{m^{-1}\ s^{-1}\ m^{\frac{1}{2}}}$

- $m\ \ s^{-1} = m^{-\frac{1}{2}} s^{-1}$

**B** is incorrect because:

- $m\ \ s^{-1} = \frac{(m\ \ s^{-2})(m)(kg\ \ m^{-3})}{kg\ \ m^{-1}\ \ s^{-1}}$
- $m\ \ s^{-1} = m^{-1}\ \ \ s^{-1}\ \ \ m$
- $m\ \ s^{-1} = \ \ \ s^{-1}$

**D** is incorrect because:

- $m\ \ s^{-1} = \frac{(m\ \ s^{-2})(m^3)(kg\ \ m^{-3})}{kg\ \ m^{-1}\ s^{-1}}$
- $m\ \ s^{-1} = m^{-1}\ s^{-1}\ m^3$
- $m\ \ s^{-1} = m^2\ s^{-1}$

Make sure to substitute the units of all of the terms into the equations. Then set *n* such that the equation is homogeneous. An equation is homogeneous when the units on both sides of the equation are the same.

**Question 22** Hard [1 mark]

The current $I$ through a semiconductor diode is given by:

$$I = I_0 e^{\frac{-qV}{nT}}$$

where $I_0$ is the initial current, $q$ is the charge of an electron, $V$ is the applied potential difference, $T$ is the temperature, and $n$ is a constant.

What is the unit of $n$ when expressed in SI base units?

**A** kg $m^2$ $s^{-2}$ $K^{-1}$

**B** kg $m^2$ $s^{-4}$ $K^{-1}$

**C** $kg^{-1}$ $m^{-2}$ $s^2$ K

**D** $kg^{-1}$ $m^{-2}$ $s^4$ K

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- From the given relationship $I = I_0 e^{\frac{-qV}{nT}}$
  - $I$ and $I_0$ are both currents with units of amperes, A
  - The exponent $\frac{qV}{nT}$ must be dimensionless (have no units)
  - Therefore, the units of $n$ must cancel the units of $\frac{qV}{T}$
- The unit of charge $q$ is the coulomb C, which in SI base units is:
  - $q = I \times t$
  - $q = \mathrm{A\ s}$
- The unit of potential difference $V$ is the volt V, which in SI base units is:
  - $V = \frac{W}{q}$
  - the unit of work done is $W = \mathrm{J} = \mathrm{kg\ m^2\ s^{-2}}$
  - $V = \mathrm{J\ C^{-1}}$
  - $V = \mathrm{kg\ m^2\ s^{-2}\ A\ s^{-1}}$
  - $V = \mathrm{kg\ m^2\ s^{-3}\ A^{-1}}$
- The unit of temperature $T$ is the Kelvin K, which is an SI base unit

- Therefore, the unit of $n$ is:
  - $n \propto \frac{qV}{l}$
  - $n = \mathrm{A\ s\ kg\ m^2\ s^{-3}\ A^{-1}\ K^{-1}}$
  - $n = \mathrm{kg\ m^2\ s^{-2}\ K^{-1}}$